

A Proofs of results

A.1 Proof of Theorem 1

Proof. Suppose that the joint distribution of (X, Y) admits the two representations

$$g(Y) = h(f_1(X_1), \dots, f_p(X_p)) + \varepsilon$$

and

$$\tilde{g}(Y) = \tilde{h}(\tilde{f}_1(X_1), \dots, \tilde{f}_p(X_p)) + \tilde{\varepsilon},$$

where

$$E(\varepsilon \mid X) = 0, \quad E(\tilde{\varepsilon} \mid X) = 0.$$

Taking conditional expectations with respect to X gives

$$E\{g(Y) \mid X\} = h(f_1(X_1), \dots, f_p(X_p))$$

and

$$E\{\tilde{g}(Y) \mid X\} = \tilde{h}(\tilde{f}_1(X_1), \dots, \tilde{f}_p(X_p)).$$

Since both representations describe the same conditional distribution of Y given X , the two transformed conditional regression surfaces are related through the monotone transformation

$$\phi = \tilde{g} \circ g^{-1}.$$

Because both g and $\tilde{g}$ are strictly increasing, ϕ is also strictly increasing. Consequently,

$$\tilde{h}(\tilde{f}_1(X_1), \dots, \tilde{f}_p(X_p)) = \phi(h(f_1(X_1), \dots, f_p(X_p)))$$

almost surely.

The centering restrictions

$$E\{f_j(X_j)\} = E\{\tilde{f}_j(X_j)\} = 0, \quad j = 1, \dots, p,$$

eliminate additive shifts in the component transformations. Similarly, the scale restrictions

$$E[h^2(f_1(X_1), \dots, f_p(X_p))] = 1$$

and

$$E[\tilde{h}^2(\tilde{f}_1(X_1), \dots, \tilde{f}_p(X_p))] = 1$$

eliminate multiplicative rescaling of the regression surface.

Under Assumption I(ii), the regression surface is one-to-one on the range of the transformed covariates. Hence, after imposing the centering and scale normalizations, the representation of the conditional regression surface in terms of the component transformations is unique. Therefore,

$$f_j(X_j) = \tilde{f}_j(X_j), \quad j = 1, \dots, p,$$

almost surely.

Substituting these equalities into the two representations gives

$$h(f_1(X_1), \dots, f_p(X_p)) = \tilde{h}(f_1(X_1), \dots, f_p(X_p))$$

almost surely. Thus,

$$h = \tilde{h}$$

on the support of the transformed covariates.

Finally, since g and $\tilde{g}$ are strictly increasing and satisfy the stated identifying normalization, the remaining monotone transformation of the response scale is uniquely determined. Hence,

$$g = \tilde{g}$$

almost surely.

Therefore,

$$g = \tilde{g}, \quad h = \tilde{h}, \quad f_j = \tilde{f}_j, \quad j = 1, \dots, p,$$

almost surely. Hence the functional transformation regression model is identifiable. $\square$

A.2 Proof of Theorem 2

Proof. For each $j = 1, \dots, p$, consider the infeasible oracle estimator

$$\hat{f}_j^o(x) = \frac{\sum_{i=1}^n K_{h_j}(X_{ij} - x) \{g(Y_i) - \mu_{-j}(X_{i,-j})\}}{\sum_{i=1}^n K_{h_j}(X_{ij} - x)},$$

where $\mu_{-j}(X_{i,-j})$ denotes the contribution of the remaining covariates. By the standard uniform convergence results for the Nadaraya–Watson kernel estimator and Assumptions A1–A4,

$$\sup_x \left| \hat{f}_j^o(x) - f_j(x) \right| = O_p(h_j^2 + (nh_j)^{-1/2}).$$

The feasible estimator $\hat{f}_j$ is obtained by replacing the unknown quantities in the oracle estimator by their current estimates. Hence,

$$\hat{f}_j(x) - f_j(x) = \left\{ \hat{f}_j(x) - \hat{f}_j^o(x) \right\} + \left\{ \hat{f}_j^o(x) - f_j(x) \right\}.$$

By Assumptions A5–A7, the effect of estimating the response transformation and the remaining component functions is uniformly negligible relative to the kernel estimation error. In particular,

$$\sup_x \left| \hat{f}_j(x) - \hat{f}_j^o(x) \right| = o_p(h_j^2 + (nh_j)^{-1/2}).$$

Therefore, by the triangle inequality,

$$\sup_x |\hat{f}_j(x) - f_j(x)| = O_p(h_j^2 + (nh_j)^{-1/2}), \quad j = 1, \dots, p.$$

Next, let $\hat{h}^o(z)$ denote the corresponding oracle estimator of $h(z)$ based on the true transformed covariates

$$Z_i = (f_1(X_{i1}), \dots, f_p(X_{ip}))^\top.$$

For the multivariate kernel estimator with bandwidth matrix H , the standard uniform kernel estimation result gives

$$\sup_z \left| \hat{h}^o(z) - h(z) \right| = O_p(|H|^{1/2} + (n|H|)^{-1/2}),$$

where $|H|$ denotes the determinant of the bandwidth matrix. The first term corresponds to the uniform smoothing bias, while the second term represents the stochastic variation of the estimator.

The feasible estimator $\hat{h}$ uses the estimated transformed covariates

$$\hat{Z}_i = (\hat{f}_1(X_{i1}), \dots, \hat{f}_p(X_{ip}))^\top.$$

Thus,

$$\hat{h}(z) - h(z) = \left\{ \hat{h}(z) - \hat{h}^o(z) \right\} + \left\{ \hat{h}^o(z) - h(z) \right\}.$$

By the uniform consistency of $\hat{f}_j$ established above, together with the smoothness and boundedness conditions in Assumptions A5–A7, the perturbation caused by replacing Z_i by $\hat{Z}_i$ is of smaller order than the oracle estimation error. Hence,

$$\sup_z \left| \hat{h}(z) - \hat{h}^o(z) \right| = o_p(|H|^{1/2} + (n|H|)^{-1/2}).$$

Combining the preceding two displays yields

$$\sup_z |\hat{h}(z) - h(z)| = O_p(|H|^{1/2} + (n|H|)^{-1/2}).$$

□

A.3 Proof of Theorem 3

Proof. The proof is first established for the oracle estimators and then transferred to the feasible estimators by showing that the effect of estimating the transformation functions is asymptotically negligible.

For a fixed $j \in \{1, \dots, p\}$ and an interior point x , consider the oracle estimator $\hat{f}_j^o(x)$ obtained by replacing the estimated transformed covariates and the estimated response transformation by their true counterparts. Under Assumptions A1–A4, the oracle estimator admits the usual local-kernel representation. A second-order Taylor expansion of f_j around x gives

$$f_j(X_{ij}) = f_j(x) + f_j'(x)(X_{ij} - x) + \frac{1}{2}f_j''(x)(X_{ij} - x)^2 + o((X_{ij} - x)^2).$$

Since the kernel is symmetric and has finite second moment, the first-order term vanishes after kernel averaging. Consequently, the leading bias of the oracle estimator is

$$\text{Bias}_{f_j}(x) = \frac{1}{2}h_j^2 f_j''(x)\mu_2(K), \quad \mu_2(K) = \int u^2 K(u) du.$$

The stochastic component can be written, up to an asymptotically negligible remainder, as a normalized weighted sum of the corresponding oracle residuals. By Assumptions A1–A4 and the bandwidth conditions in A7, the Lindeberg condition is satisfied. Hence, the Lindeberg–Feller central limit theorem yields

$$\sqrt{nh_j} \left\{ \hat{f}_j^o(x) - f_j(x) - \text{Bias}_{f_j}(x) \right\} \xrightarrow{d} N\left(0, \sigma_{f_j}^2(x)\right).$$

The variance $\sigma_{f_j}^2(x)$ is the usual pointwise asymptotic variance determined by the conditional variance of the oracle residual and the kernel factor.

It remains to show that replacing the unknown transformation functions by their feasible estimates does not alter the limiting distribution. By the uniform consistency results established under Assumptions A5–A7,

$$\sup_x \left| \hat{f}_j(x) - \hat{f}_j^o(x) \right| = o_p((nh_j)^{-1/2}).$$

Therefore,

$$\sqrt{nh_j} \left\{ \hat{f}_j(x) - \hat{f}_j^o(x) \right\} = o_p(1).$$

Combining this result with the oracle central limit theorem and applying Slutsky’s theorem gives

$$\sqrt{nh_j} \left(\hat{f}_j(x) - f_j(x) - \text{Bias}_{f_j}(x) \right) \xrightarrow{d} N\left(0, \sigma_{f_j}^2(x)\right).$$

Next, consider the multivariate regression surface $h(z)$, where z is an interior point of the support of the transformed covariate vector. Let $\hat{h}^o(z)$ denote the corresponding oracle estimator constructed from the true transformed covariates

$$Z_i = (f_1(X_{i1}), \dots, f_p(X_{ip}))^\top.$$

Under Assumptions A1–A4 and the smoothness conditions in A5–A7, a second-order multivariate Taylor expansion of h around z yields the standard kernel bias expansion

$$\text{Bias}_h(z) = \frac{1}{2} \sum_{k=1}^p \sum_{\ell=1}^p H_{k\ell} \frac{\partial^2 h(z)}{\partial z_k \partial z_\ell} \mu_{k\ell}(K) + o(\|H\|),$$

where

$$\mu_{k\ell}(K) = \int u_k u_\ell K(u) du.$$

Thus, after centering by the leading bias, the oracle estimator has the representation

$$\sqrt{n|H|} \left\{ \hat{h}^o(z) - h(z) - \text{Bias}_h(z) \right\} = \frac{1}{\sqrt{n|H|}} \sum_{i=1}^n W_{ni}(z) \varepsilon_i + o_p(1),$$

where $W_{ni}(z)$ denotes the corresponding normalized kernel weight. The kernel assumptions and bandwidth conditions in A7 ensure that the maximum contribution of an individual

summand is asymptotically negligible and that the conditional variance converges to a finite, positive limit. Hence, another application of the Lindeberg–Feller central limit theorem gives

$$\sqrt{n|H|} \left\{ \hat{h}^o(z) - h(z) - \text{Bias}_h(z) \right\} \xrightarrow{d} N(0, \sigma_h^2(z)) .$$

Finally, the feasible estimator $\hat{h}(z)$ is obtained by replacing the true transformed covariates Z_i by

$$\hat{Z}_i = (\hat{f}_1(X_{i1}), \dots, \hat{f}_p(X_{ip}))^\top .$$

The uniform consistency of the component estimators, together with the smoothness and boundedness conditions in Assumptions A5–A7, implies

$$\sup_z \left| \hat{h}(z) - \hat{h}^o(z) \right| = o_p((n|H|)^{-1/2}) .$$

Consequently,

$$\sqrt{n|H|} \left\{ \hat{h}(z) - \hat{h}^o(z) \right\} = o_p(1) .$$

Combining this result with the oracle asymptotic normality and applying Slutsky’s theorem gives

$$\sqrt{n|H|} \left(\hat{h}(z) - h(z) - \text{Bias}_h(z) \right) \xrightarrow{d} N(0, \sigma_h^2(z)) .$$

The proof is therefore complete. □